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To cite this article: Kisung Jeong, Jiwon Lee & Jinmo Kim (2017): A Study on New Virtual Reality System in Maze Terrain, International Journal of Human-Computer Interaction, DOI: [10.1080/10447318.2017.1331535](https://doi.org/10.1080/10447318.2017.1331535)

To link to this article: <http://dx.doi.org/10.1080/10447318.2017.1331535>



Accepted author version posted online: 16 May 2017.
Published online: 16 May 2017.



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A Study on New Virtual Reality System in Maze Terrain

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ABSTRACT

This study proposes the maze virtual reality (MVR) system, which is designed to provide new immersive virtual reality (VR) experiences by generating a maze and an event-based maze environment. The MVR system consists of a maze generation technique based on mazemap, which we additionally propose for the automatic generation of diverse maze patterns, event-enhancing immersion, reducing VR sickness, and interactions. First, we present a finite algorithm that automatically generates diverse maze patterns from the user perspective and implements mazemap for automated three-dimensional (3D) terrain generation from a 2D maze pattern. Then, we present optimized events for the maze environment by providing users with deeper immersion and minimized VR sickness. The design procedure is divided into two elements: an automatic structure for arranging predefined events based on maze pattern analysis and a manual operation module. In addition, a Leap Motion-based interactive scheme is implemented for deeper immersion when the user moves in a limited space, while improving object control. Finally, experiments were conducted to analyze user satisfaction with the MVR system, the psychological impact of the continuous mazemap input, and the effects of the event-based environment on immersion enhancement and VR sickness reduction.

1. Introduction

Virtual reality (VR) refers to a specific environment or situation that is artificially created by computer technology or another means to mimic the real environment or situation. This type of virtual environment or situation is generated using devices, such as a head-mounted display (HMD). It provides users with spatial and temporal experiences that are similar to reality by satisfying visual sensations and stimulating tactile, auditory, and other senses. Users experience not only a mere immersion in the VR environment through visual information, but also interactions with the virtual objects or situations by manipulating or commanding those objects using VR devices. The components that are required for implementing virtual environments include three-dimensional (3D) spatiality, real-time interaction, immersion, etc. Recent advancements in VR-related technologies, such as motion capture devices, Leap Motion, data gloves, and treadmill-type devices (e.g., Virtuix Omni), have expanded 3D spatiality, thereby diversifying interactive experiences and enhancing the sense of immersion.

Research on HMD systems that are designed to satisfy user visual sensations began in the early 1960s (Sutherland, 1968). Auditory and haptic elements were gradually added to VR-related studies to improve user realism and immersion in VR environments by inducing physical haptic feedback (Bergamasco, 1993; Brooks, 1999). More recently, studies involving prop-based VR systems have been conducted with the aim of enriching spatial experiences and delivering novel

VR scenes (Cheng, Roumen, et al., 2015). Nonetheless, although these research efforts have enabled the implementation of immersive physical actions and reactions involving moving, touching, and grasping in a given VR space, it remains a significant challenge to automatically generate new VR scenes and spaces for each situation. In this regard, a maze has advantages over nature or urban environments with forests or buildings because the maze has a simple and unique structure for delivering new environments to foster immersion.

Based on the above context, we herein present the maze virtual reality (MVR) system, which uses an automated method for maze generation from the user perspective. It enables unlimited generation of new experiences in an immersive VR environment. This article on MVR system provides three key contributions:

- A novel method for automatically generating a new immersive virtual maze pattern in each new situation by implementing mazemap by using a maze generation algorithm.
- Deeper immersion and minimized VR sickness in a virtual maze environment by automatic event generation and interaction processes that are tailored to maze environments.
- Systematic testing and analysis of the psychological effects of the proposed VR scenes and interactions, including immersion and sickness.

An MVR system comprises the mazemap-based terrain and immersive events. It is a user-oriented system that differentiates information between automatic generation and manual configuration. It thereby enables applications in a variety of fields.

The remainder of this article is organized as follows. [Section 2](#) presents earlier related studies on VR technologies and maze terrains. [Section 3](#) explains the implementation of mazemap using a maze generation algorithm and a height-map. [Section 4](#) outlines the design process of immersive events and interactions that are tailored to a maze environment. [Section 5](#) describes the procedures for maze generation and the proposed events and interactions. It additionally discusses the experiments that were conducted to analyze the psychological effects of the proposed system in terms of user immersion and VR sickness. [Section 6](#) explains the limitations and discussion. [Section 7](#) presents our conclusions and suggests interesting directions for further study.

2. Previous Work

Sutherland (Sutherland, 1968) presented research on a head-mounted VR system for transferring 3D visual information that is conducive to VR environments. This pioneering study was followed by research on haptic systems, motion platforms, and flight simulations that incorporate additional sensations, such as auditory and tactile sensations (Burdea et al., 1996; Cheng, Lühne, Lopes, Sterz, & Baudisch, 2014; Hoffman, 1998; Kohli, Burns, Miller, & Fuchs, 2005). In addition, Lopes et al. (Lopes & Baudisch, 2013) proposed a force feedback system that can induce interactions using electrical muscle simulations, while Yano et al. (Yano, Miyamoto, & Iwata, 2009) designed a handheld haptic device that enables the user to touch virtual objects using fingers. Laycock et al. (Laycock & Day, 2003) conducted a study on enhancing realism in virtual environments by enabling interactions for object control based on haptic devices. Furthermore, Danieau et al. (Danieau et al., 2012) presented the HapSeat system, which is designed to run low-cost simulations of computation steps for user head and hand motions. Moreover, Li et al. (Li et al., 2015) conducted a study on facial performance with an HMD with a sensor attached to it. Using this system, the HMD-embedded sensor and an external Red/Green/Blue-Depth (RGB-D) camera facilitated accurate measurements of facial movements, and expressed them in a virtual environment. Furthermore, motion simulators were studied to induce shaking, lifting, and tilting movements, or to control sitting and standing postures, in order to offer multifaceted experiences in VR environments by reducing spatial constraints (NASA, 2008). Driving and flight simulation environments were used for training and teaching purposes. A wearable machine enables both physical movements and force feedback. It can control all joint motions of the body with realistic accuracy, thus enabling natural movements in a VR environment (Hayashi, Kawamoto, & Sankai, 2005). Spatial constraints were progressively resolved with the development of systems that measure user motions and behaviors with notable precision and reflect them in the VR environment with a high degree of visual realism. Accordingly, physical props have gained importance as elements for reproducing virtual spaces.

In addition, Ortega et al. (Ortega & Coquillart, 2005) proposed a system that enables the user to hold a prop using a hand, and Cheng et al. (Cheng, Roumen, et al., 2015) conducted a study on interactions with physical objects via motions of holding or opening a door or pushing a wall. Their approach employs various props, such as walls, doors, ledges, and steps. However, most of the above-mentioned studies focused on hardware for addressing the user's haptic senses or motions. Studies on enabling the creative expression of visual information and other senses remain lacking. Moreover, most of the devices and systems implemented as a result of the above-mentioned studies are valuable as research; however, their diffusion among general users has not been achieved.

In recent years, Leap Motion has emerged. It enables the recognition of finger movements and motions to facilitate free 3D motion of fingers. This technology can be easily applied on account of its relatively low costs and familiar market infrastructure. Related studies have expanded technologies for interactions in virtual environments. Furthermore, studies were conducted to develop signature or authentication systems and their verifications by capturing handwriting motions through the recognition of user hand gestures and fingertip coordinates (Chan, Halevi, & Memon 2015; Kamaishi & Uda, 2016). The results of these studies have shown that the approaches are applicable to various fields, such as surgical experiments for medical training purposes (Juanes, G'omez, Peguero, & Ruisoto, 2015) and games using hand motions as interfaces (Lee, Wang, Tung, Lin, & Valstar, 2015). However, not many studies have explored their potentials for applications to interactions in VR spaces.

Analyzing the psychological factors associated with VR technologies, such as VR sickness, is as important as analyzing the factors for immersion enhancement. One study involved a systematic analysis of psychological factors to understand the factors inducing motion sickness and categorized them into personal traits, such as age, and into content-related technological elements, such as frames per second (fps) rate and the interface (Kolasinski, 1995; Moss & Muth, 2011). Other studies investigated the correlations between motion sickness and cue conflict (Duh, Parker, Philips, & Furness, 2004) or posture (Murata, 2004; Stoffregen & James, 1998). These studies demonstrated that both immersion and sickness should be considered when aiming to provide users with new immersive VR experiences.

To enable these new experiences in an approach that differs from those in existing VR-related studies, we propose the MVR system, which consists of maze-based VR scenes and interactive methods. We analyzed the proposed system in terms of immersion and sickness by conducting experiments, and the psychological effects were investigated. The maze generation algorithm of the proposed system generates maze patterns with road and wall elements by recursively processing the mazemap arrangement in cell units. In many earlier studies, a road was selected with a depth-first search algorithm and the surrounding walls were then created. Later, Prim's algorithm was studied in which the roads and walls were generated by recursively exploring neighboring cells on the basis of a minimum spanning tree (Prim, 1957). Other algorithms include Kruskal's algorithm (Kruskal, 1956), in

which mazes are generated with roads and walls arranged with randomly selected cells. Storage space is assigned that is proportional to the size of the maze, and the edges are sorted in a preset order. A study was conducted on the generation of organic labyrinths and mazes that combined the interactive artistic control of the non-photorealistic rendering (NPR) algorithm with a curve drawing based on a maze-generated algorithm (Pedersen & Singh, 2006). A study was also proposed for the automatic generation of unique maze pattern images by receiving a path from the user's starting point to the ending point (Xu & Kaplan, 2007). In addition, one study used maze patterns as interactive art techniques (Cheng, Xiang, Lyn, & Thang, 2015). However, apparently no study has been conducted on a terrain-based VR system using existing maze generation algorithms. In addition, Lai et al. (Lai, Yeung, Yan, Fu, & Tang, 2016) dynamically generated 3D mazes from impossible figures and suggested 3D navigation for guiding the path. This provided interaction with seamless gaming navigation using a novel calculation method that generates 3D structures from guiding principles and impossible figure images. This maze was reconfigurable for 3D games but did not account for factors such as presence and VR sickness that must be examined for application in VR. Therefore, the present study proposes a novel mazemap method that combines a maze generation algorithm with a heightmap as a means to effectively apply maze terrains in VR.

3. Mazemap-Based Maze Terrain

When generating new maze-based VR scenes, it is important to ensure the diversity of maze patterns and the efficiency of terrain generation. High costs and long timeframes are required for a developer to create complex maze patterns and 3D terrains comprising precise paths by repeating manual processing for each maze terrain scene. To address this problem, the proposed mazemap can efficiently generate 3D maze terrains by calculating various maze patterns and processing them with heightmap through a maze generation algorithm. The generated terrain is given a more natural appearance with the aid of the terrain editing function of

Unity3D. The mazemap-generated maze terrain is designed by an automated process. The work for adding more realism to the generated maze terrain using texture, grass, trees, and props is manually performed because it requires a graphic sense. Figure 1 outlines the proposed mazemap-based terrain generation method.

3.1. Maze Generation Algorithm

The first step of mazemap-based maze terrain generation is the computation of various maze patterns of any size desired by the user. In this process, special care should be taken to prevent the user from entering an infinite loop of maze paths because a long confinement in a maze can lead to motion sickness and dizziness. Therefore, the maze generation algorithm should be designed to generate finite mazes that satisfy the following two conditions:

- The maze is noncyclical to enable only one path from the entrance to the exit.
- In the maze, no closed wall exists to block the user's entrance.

In this study, the mazes that fulfill these two conditions are generated using Prim's algorithm, which computes the minimum spanning tree. Accordingly, the lowest-cost spanning tree is obtained by beginning with the start node and progressively expanding the set of spanning trees. A finite maze can be generated by applying the procedure for searching the lowest-cost path to the maze generation. Other algorithms exist for generating finite mazes, including Kruskal's and growing tree algorithm. However, this study transcends mere maze pattern generation; it additionally designs immersive VR environments by applying the generated mazes to any given VR environment. Prim's algorithm is a greedy algorithm that initializes a node and finds the final minimum weight. Thus, it is suited to the objective of this study, which presupposes clear start and end positions. To this end, the original Prim's algorithm was modified to better align with the user-perspective VR environment by generating

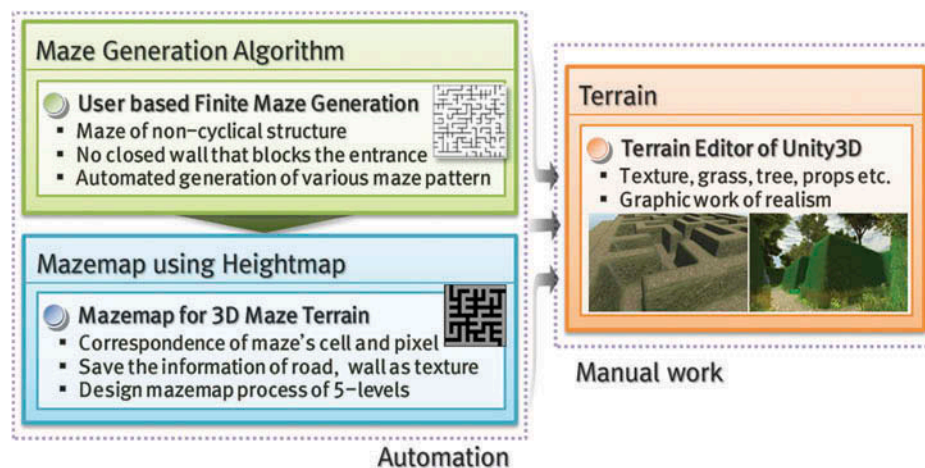


Figure 1. Overview of mazemap-based terrain generation.

a maze starting from a predetermined position and by preventing infinite looping before the end position.

Prim's algorithm-based maze pattern generation follows the steps described below. The maze is sized in accordance with the user's desire. The smallest unit of the maze is defined as a cell, and each cell is used for either a road or a wall.

- (1) The starting cell is selected. In this study, the start and end positions were set to be the center and the four corners of the maze, respectively. This arrangement has the advantages of shortening the time the user remains in the maze by setting multiple destinations from the center and diversifying the experience of escaping the maze.
- (2) One of the four cells neighboring the start cell is randomly chosen. The selected cell (p_{cell}^{road}) is set as a road.
- (3) The four cells surrounding the selected road cell are set as walls, except for a wall cell, which is already defined as a road cell. The road cell (p_{cell}^{road}) becomes the parent cell (p_{cell}^{parent}) of the wall cells.
- (4) A randomly selected wall cell is set as a candidate cell (p_{cell}^{cand}), and it is used for deciding the next road. This candidate cell is excluded from the next set of wall cells.
- (5) The next road cell ($p_{cell}^{road} = p_{cell}^{cand} + \vec{v}_{road}$) is computed by considering the direction of the path. This direction can be derived from the difference between the candidate cell and its parent cell ($\vec{v}_{road} = p_{cell}^{cand} - p_{cell}^{parent}$).

Lastly, the walls surrounding the road cell are to be computed as in Step (3).

A maze pattern is generated by sorting roads and walls while repeating Steps (4)–(6). Figure 2 illustrates the process for determining the roads and walls using the proposed maze generation algorithm and the resulting maze pattern. The sign “+” denotes a wall, and “0” denotes a road. Using these signs, it can be confirmed that all finite paths run from the center of the maze toward the respective ends of the four directions. In this study, finite mazes were generated to prevent users from falling into an infinite path in VR. However, if a cyclical maze is preferred, the process of removing the candidate cells from the set of wall cells in step (3) can be omitted, and the process of finding the road cell can be repeated.

3.2. Mazemap Using Heightmap

To express a computed finite maze pattern as a 3D graphic in the VR space, it is necessary to reproduce it in the corresponding maze. A common approach to this implementation is to simply arrange the cells to match their corresponding props. However, only one maze pattern is formed in this way; it thus cannot fulfill the function of offering new VR experiences. To address this drawback, we propose mazemap, which expresses the computed maze pattern as a natural terrain to enhance realism and immersion. In general, 3D interactive contents, such as game terrains, are generated using heightmap. Complex terrains can be reproduced by simply converting the grayscale stored in a heightmap into the height value. Mazemap was designed to store the road

and wall height information as textures by matching the road and wall cells of the generated finite maze with their respective image pixels. The procedure for mazemap consists of the five steps detailed below and outlined in Algorithm 1.

Algorithm 1. Mazemap algorithm using a heightmap

```

C[x][y] ← x x y maze cell information consisting of the road and wall.
procedure MAZEMAP (C)
  I[x][y] ← initial image of mazemap.
  M[2n][2n] ← 2n-sized mazemap
  (1) Generate an image (I) as large as the computed finite maze (C).
  (2) Inspect the external wall to prevent an open wall.
  if vertical wall is an open wall then
    Add vertical wall (y ← y + 2).
  end if
  if horizontal wall is an open wall then
    Add horizontal wall (x ← x + 2).
  end if
  (3) Set the four end points starting from the center.
  (4) Set the margins with consideration of the scenery.
  x ← x + margin * 2
  y ← y + margin * 2
  (5) Generate a heightmap that matches the relative size of the maze with
  a 2n texture.
  return M (← heightmap)
end procedure

```

(1) An image of the same size as the computed maze is generated and stored in the values of 0 and 128 for roads and walls, respectively (Figure 3(a)). The value of 128 for the wall height is the median value for 8 bits/pixel. (2) The external walls are inspected (left and right, up and down). Not all computed maze patterns have external walls (Figure 3(b)). If any external wall is missing, it is added because it is difficult to set the destination without all external walls in place. (3) Four end points are set at the left top, left bottom, right top, and right bottom corners from the center point. By setting four different end points, the user can escape faster and experience four different escape routes. The four end points are identified and walls are broken through with mazemap generation in Steps (1) and (2) (Figure 3(c)). (4) The margins are added to mazemap at regular intervals in proportion to the maze pattern size to accommodate surrounding landscapes in addition to the maze. (5) The heightmap reflecting the size of the maze pattern is converted to a 2ⁿ texture, whereby the first concern is the mazemap resolution for the maze pattern. If the resolution is fixed, the maze pattern size determines the road and wall widths. That is, if the maze pattern size exceeds the standard value, the road can become too narrow for the user to move along (Figure 3(d)). Thus, a mazemap is generated in proportion to the maze pattern size (Figure 3(e)). The second concern is the resolution of the heightmap, which is usually set at 2ⁿ. The computed mazemap is thus finally generated in a 2ⁿ texture (Figure 3(f)). This procedure enables the automatic generation of diverse mazemaps of adequate widths.

In addition to illustrating the generation of a mazemap, Figure 3 depicts the problems that may be encountered while generating a mazemap, as well as the resulting mazemap after the problems are addressed. As shown in Figure 3(g), mazemap-based terrain generation yields terrains that have road–wall

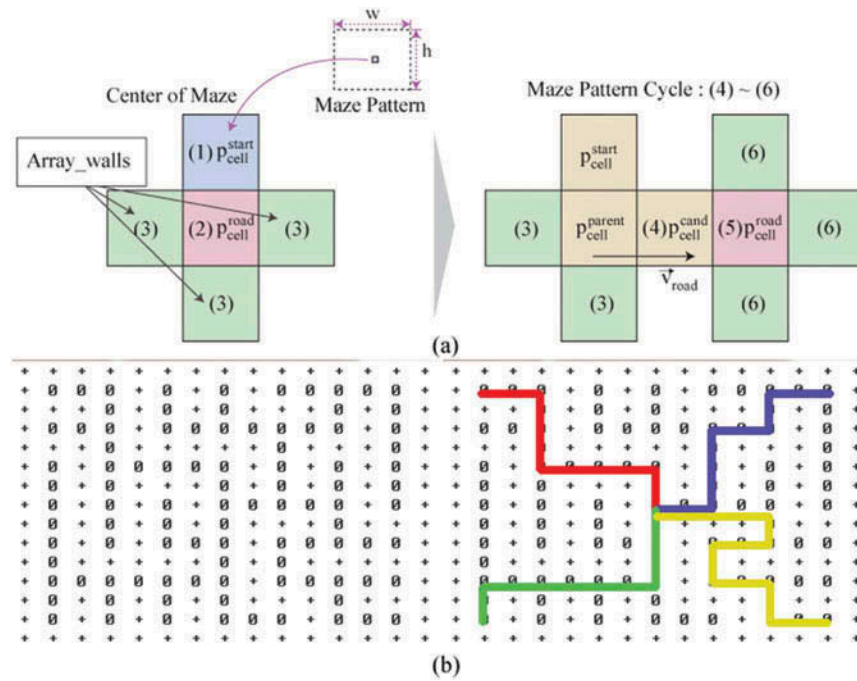


Figure 2. Finite maze patterns generated with the proposed maze generation algorithm. (a) Computing process of the finite maze generation algorithm. (b) Resulting finite maze patterns running from the center point toward four destinations.

boundaries that are too clearly delineated and thereby appear artificial. Natural boundaries were thus implemented using a Gaussian blur.

Mazemap additionally generates non-square maps, as shown in Figure 4. In this study, however, mazemap was designed to generate mazes that are as square as possible with consideration of the balance of options in the procedure and time for escaping in the four destinations.

3.3. 3D Terrain

Once the shape of a 3D maze terrain is generated using mazemap with diverse maze patterns, maze images should undergo a process to enhance realism. Scenes in a maze terrain of the same pattern vary according to the purpose of the contents to be produced. The working process of completing the maze terrain using texture and props, which involves a high graphical sense, cannot be automatically generated. Therefore, this process of producing a mazemap-based nature-like terrain is manually performed.

We used the Unity3D terrain editor for efficient terrain production (Figure 5). Unity3D was selected because it offers an integrated environment that accounts for interoperability with other VR devices, such as the Oculus Rift HMD and Leap Motion SDK. The terrain design is prepared with premade texture, grass, trees, and props. It is difficult to generally estimate the time requirement for the manual work on the terrain because it varies according to the expertise and design sense of the worker. For this study, it required 15 hours to generate a terrain with a maze pattern (27×27) and to manually complete it in the state shown in Figure 5. It took about 6–7 hours to draw the 27×27 pattern maze shown in Figure 3 and to manually create a 3D terrain maze (Figure 5 (a)) using the Unity 3D terrain editor. When the proposed algorithm is used, the maze pattern can be generated

automatically within 1 minute, and the 3D maze terrain can be completed in around 5 minutes. Thus, the proposed algorithm can save one-third of the total manual work time. The proficiency of the graphic designer who participated in this experiment was limited to program development. Even if the graphic designer's ability to use the proposed algorithm is compared with the ability of a more skilled designer, who has a better graphic sense, it would still take much less time to complete the total maze terrain work, including graphic work, using the proposed algorithm.

4. Immersive Events and Interaction

The mazemap-generated terrain can independently offer users new VR experiences. However, the presence of event elements that enhance tension would make the VR experiences more immersive for users. More importantly, a long confinement in a maze may cause fatigue and sickness to the user, especially in a VR. The proposed MVR system was designed to incorporate events and interactions that are tailored to a VR environment by enhancing user immersion and minimizing VR sickness. The MVR system designing of the proposed events is conducted in both automated and manual processes that account for immersion and safety in a VR environment. The key function of the MVR system is to provide a method to automatically select and arrange events that are optimized for maze environments by analyzing the maze pattern. Lastly, immersion is increased by interactions that are implemented using hand motions based on Leap Motion.

4.1. Maze-Appropriate Events

We designed events that are optimized for maze environments in two categories: immersion-enhancing events and sickness-

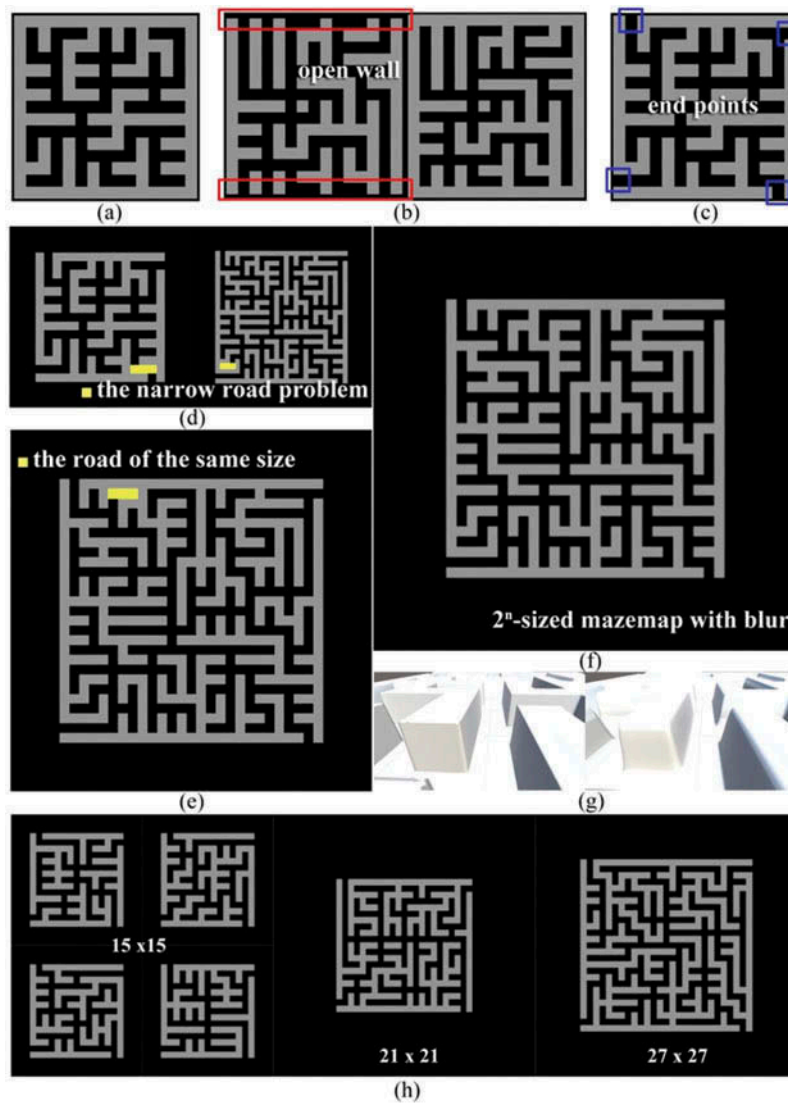


Figure 3. Procedure for generating a mazemap using a heightmap: (a) generating an image of the same size as the computed maze, (b) problem-solving through external wall inspection, (c) determining the four end points, (d) setting margins by considering the scenery and spatial issues and adjusting the resolution, (e) generating a mazemap that reflects the maze size, (f) converting it into a 2^n texture and applying a Gaussian blur, (g) result of smoothing the terrain image with a Gaussian blur, (h) azemaps with diverse maze patterns.

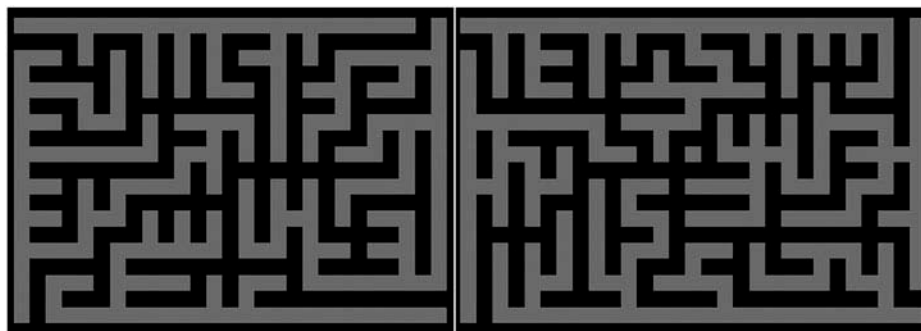


Figure 4. Results of non-square mazemap generation.

reducing events. An immersion-enhancing event is defined as an automatically generated event based on the analysis of the maze-map maze pattern. Event elements are designed in three categories: animals, props, and particles, as illustrated in Figure 6. During the

maze experience, these elements suddenly appear to frighten the user while he/she is concentrating on escaping the maze.

Maze pattern templates are defined to enable the proposed events to be automatically matched to the corresponding

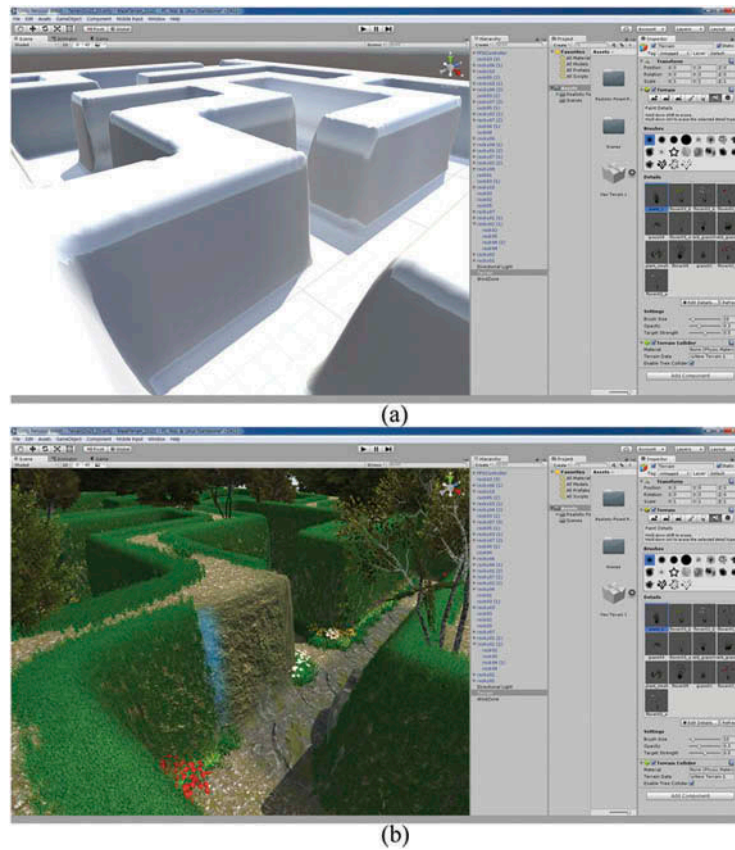


Figure 5. Terrain production using Unity3D: (a) mazemap-generated maze terrain, (b) Manual graphic work using texture, grass, trees, and props to enhance realism.

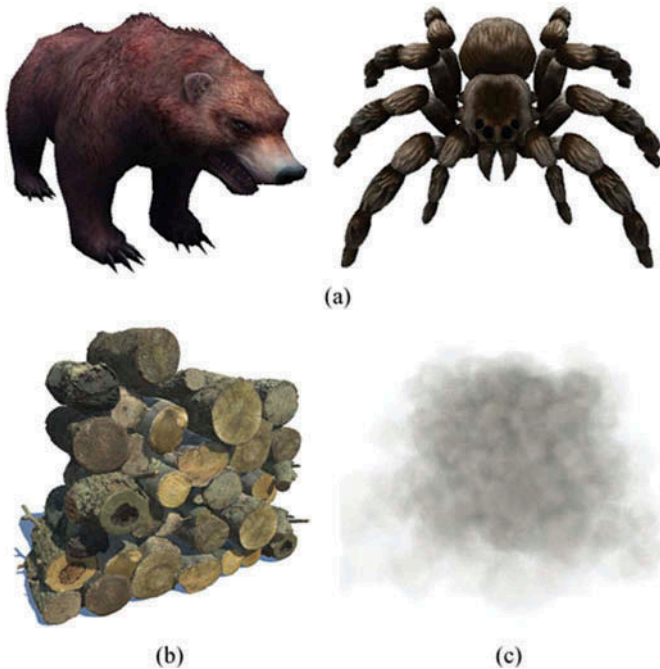


Figure 6. Event elements used for enhancing immersion: (a) animals, (b) props, (c) fog particles.

maze patterns. Figure 7(a) shows configurations for an animal that suddenly appears on a curved road. Figure 7(b) depicts props dropping from the sky while the user moves along a direct road. Figure 7(c) shows crossroads and dead ends

where particles and sounds are used to increase the user's confusion and tension. Event elements are equally placed among the roads toward the four mazemap destinations. Algorithm 2 presents the automated event generation procedure for matching event elements to maze patterns.

Once the cells for events are decided through maze pattern analysis, their precise 3D coordinates should be derived. If the cell coordinates ($c(x, y)$) of an event are automatically derived by Algorithm 2 in a maze pattern ($n \times n$), as shown in Figure 2(b), the equation for computing their 3D coordinates ($X(x, z)$) on the maze terrain ($N \times N$) is defined. First, the external wall is inspected in the mazemap with Step (2). If one or more external walls should be added, the size of the mazemap is increased by the number of detected external walls (n_{extw}). That is, n_{extw} is calculated at the process step for external wall inspection. In mazemap Step (4), margins (l_{margin}) that account for the landscape are set at the adjustable default size of $1/5$ of the maze pattern size. Finally, in Step (5) (i.e., the process of the heightmap conversion considering the relative size), the difference value is obtained. In this study, the design is based on the minimum maze pattern size of 15 and its corresponding minimum mazemap resolution of 512. That is, if the size of the maze pattern increases by k , the resolution (r_{curr}) of the mazemap increases by $(15 + k) / 15 \times 512$ accordingly. Because the resolution of the resulting mazemap does not reach the value 2^n , a reference resolution is set for converting the size of the generated mazemap into 2^n . For example, if the size of the generated mazemap is 717 ($k = 5$), the reference resolution (r_{ref}) is 1024, and the mazemap enlargement yields ΔM of $(1024 - 718) / 2$.

Algorithm 2. Automated arrangement of events to their corresponding maze patterns

$p \leftarrow$ any given cell within the maze pattern.
 $List_events[3] \leftarrow$ list with cell positions of three types of events.
 $c[8] \leftarrow$ Search cells in eight directions adjacent to $p((p.x - 1, p.y - 1) \rightarrow (p.x + 1, p.y + 1))$.
procedure EVENT_ANIMAL ($p, List_events, c$)
if two cells among the cells found (c) are road cells **then**
 $p_{cell}^{road}[2] \leftarrow$ two road cells found.
if $p_{cell}^{road}[0].x \neq p_{cell}^{road}[1].x$ **And** $p_{cell}^{road}[0].y \neq p_{cell}^{road}[1].y$ **then**
 place animal events on road cells (p_{cell}^{road}).
 $List_events[0].add(p_{cell}^{road})$.
end if
end if
end procedure
procedure EVENTS_PROPS ($p, List_events, c$)
if two cells among the cells found (c) are road cells
 $p_{cell}^{road}[2] \leftarrow$ two road cells found.
if $p.x = p_{cell}^{road}[0].x$ **And** $p.x = p_{cell}^{road}[1].x$ **then**
 place dropping props at p .
 $List_events[1].add(p)$.
else if $p.y = p_{cell}^{road}[0].y$ **And** $p.y = p_{cell}^{road}[1].y$ **then**
 place dropping props at p .
 $List_events[1].add(p)$.
end if
end if
end procedure
procedure EVENT_PARTICLES ($p, List_events, c$)
if only one cell of the cells found (c) is road **Or** only horizontal or vertical cells of p are walls **then**
if only one cell of the cells found is road **then**
 place a particle event on road cell (p_{cell}^{road}).
 $List_events[2].add(p_{cell}^{road})$.
else if only horizontal or vertical cells of p are walls **then**
 place a particle event at p .
 $List_events[2].add(p)$.
end if
end if
end procedure
 Divide $List_events$ into four areas and evenly place them on the mazemap.

Equation (1) represents the process of computing the 3D coordinates ($X(x, z)$) at the position of the event cell ($c(x, y)$).

$$w_s = \frac{n}{15} * \frac{512}{n + l_{margin} * 2 + n_{extw}},$$

$$X(x, z) = \frac{N}{r_{ref}} (w_s * c(x + l_{margin}, y + l_{margin}) + (\Delta M, \Delta M)) \quad (1)$$

Events also exist for sickness mitigation. Specifically, they are road-finding events designed to help the user find the path leading to the end point in order to reduce sickness. As mentioned earlier, the user will likely experience this sickness when confined in the maze for a prolonged length of time. There are three types of sickness-reducing events. One is placing a prop in a suitable position as a reference point if the user returns from a dead end to prevent error repetition. The second is placing a particle prop if the user passes a confusing crossroad twice. The third is providing a mini-map to help the user find the path if he/she fails to escape from the maze after the lapse of a certain length of time. In a preliminary study, 90 seconds was found to be the maximum time for the user to remain in the maze without incurring sickness. Therefore, a mini-map was configured for availability at 70 seconds to give the user 20 seconds to escape the maze using the map. Additionally, we experimentally analyzed the effects of the time spent within a maze on sickness, as described in Section 5. Figure 8 presents the sickness-reducing events for road-finding. They are events directly associated with sickness and are mandatory, unlike immersion-enhancing events for which reactions are optional. Therefore, the placement of these events is manually configured on the basis of the generated mazemap to compel the user to react to them.

4.2. Hand Interaction Using Leap Motion

Finally, we propose a Leap Motion-based interaction method that mitigates the user's spatial constraints in an MVR system and enables a high degree of free motion. The most salient features of this method are the effects of increasing the freedom of movement and minimizing motion sickness in a virtual space. These effects can be achieved by using motion capture devices that can recognize a user's gesture and posture, or treadmills, such as Virtuix Omni, which are very expensive and thus beyond

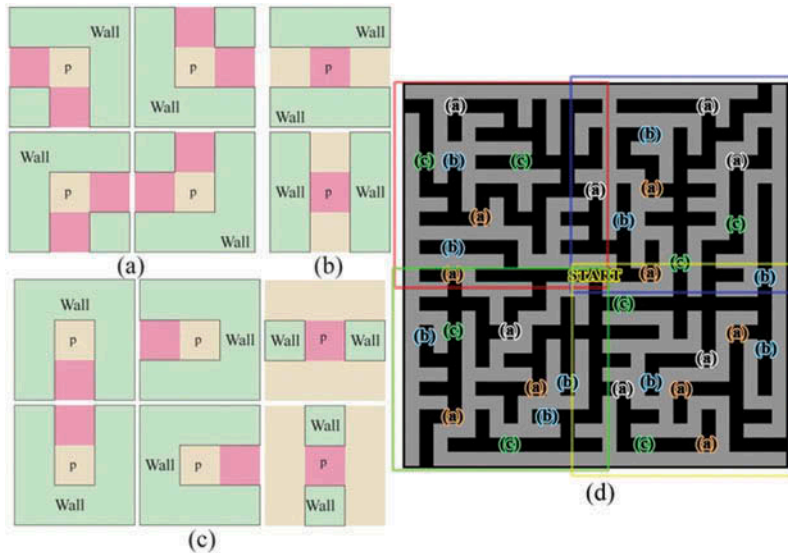


Figure 7. Maze pattern templates for automated event generation: (a) animal event template, (b) prop event template, (c) particle event template, (d) automatic arrangement of events in different maze areas.

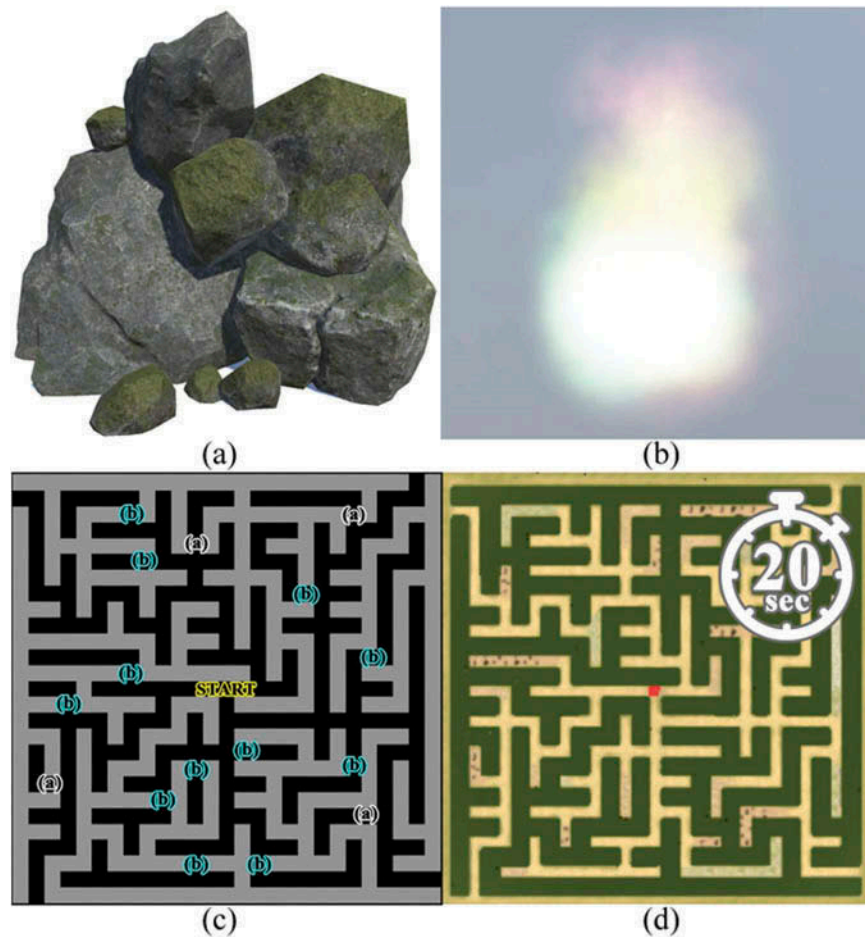


Figure 8. Sickness-reducing events for road-finding: (a) removing a dead end road, (b) signaling repeated visits to the same crossroad, (c) placing events according to the maze pattern, (d) providing path information with a mini-map.

the reach of most users. To overcome this problem, we designed Leap Motion-based interactions to directly express movement-related sensations instead of using common input devices, such as a game pad or keyboard.

The Leap Motion sensor is an instrument consisting of two infrared cameras and an LED. This infrared sensor is a small USB peripheral (12.7 mm × 80.0 mm), which is mountable to the HMD. It recognizes, stores, and processes hand motions. Our interaction design facilitates natural movements in a maze by defining the actions that the user can take in a maze in four steps. Figure 9 presents each of the four movement definitions of walking, left/right translation, stop, and object control. The object control movement allows the user to directly remove immersive events, thus enhancing user immersion.

5. Experimental Results and Analysis

The proposed mazemap was implemented using Microsoft Visual Studio 2013 and OpenCV 2.4.11. Unity3D 5.3.4f1, Oculus SDK (ovr_unity_utilities 1.3.2, OVRPlugin 1.3.2), and Leap Motion SDK v4.1.4 were used for implementing the maze-based VR environments. The PC environment used for the MVR system experiments was equipped with Intel Core i7-6700, 16 GB RAM, and a GeForce GTX 1080 GPU.

The first experiment focused on maze terrain generation. Figure 10 illustrates the procedure involved in realistically generating an MVR environment using the proposed mazemap to provide users with new experiences. First, the mazemap was generated in any size of maze pattern as input by the user for automatic generation of the maze terrain (Figures 10(a–c)). Then, the terrain was given a realistic expression by sequential manual work on the texture, grass, trees, and props (Figures 10(d) and (e)). When creating this VR content, special attention was given to the refresh rate (Hz) and frame rate (fps). The proposed virtual maze scenes were generated at 75 fps (Table 1) or higher using the level of detail (LOD) provided by terrain editor in Unity3D to prevent VR sickness caused by latency (input lag).

If the height of a maze provided by the mazemap is constant, as shown in Figure 10, it is possible to generate an organized maze environment; however, if one wants to emphasize the feeling of a natural maze, the height should be irregular. Figure 11 shows a mazemap with irregular heights (Figure 11 (b)) generated by combining a Gaussian noise map (Figure 11 (a)) with the generated mazemap (Figure 10 (a)). Then, a scene, such as the one shown in Figure 11 (c), can be provided by automatically generating a maze terrain based on this mazemap. This can be used as a natural maze terrain with a reduced artificial feeling. One important point is that the proposed MVR provides paths and events in such a manner that they are not predictable

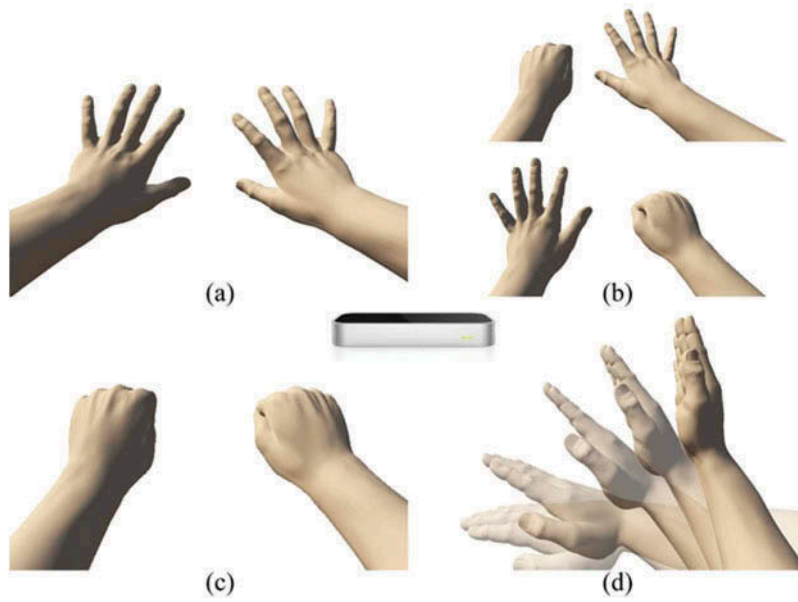


Figure 9. Four types of Leap Motion-based interactive motions: (a) forward moving, (b) moving to the right (top) or left (bottom), (c) stopping, (d) object control.

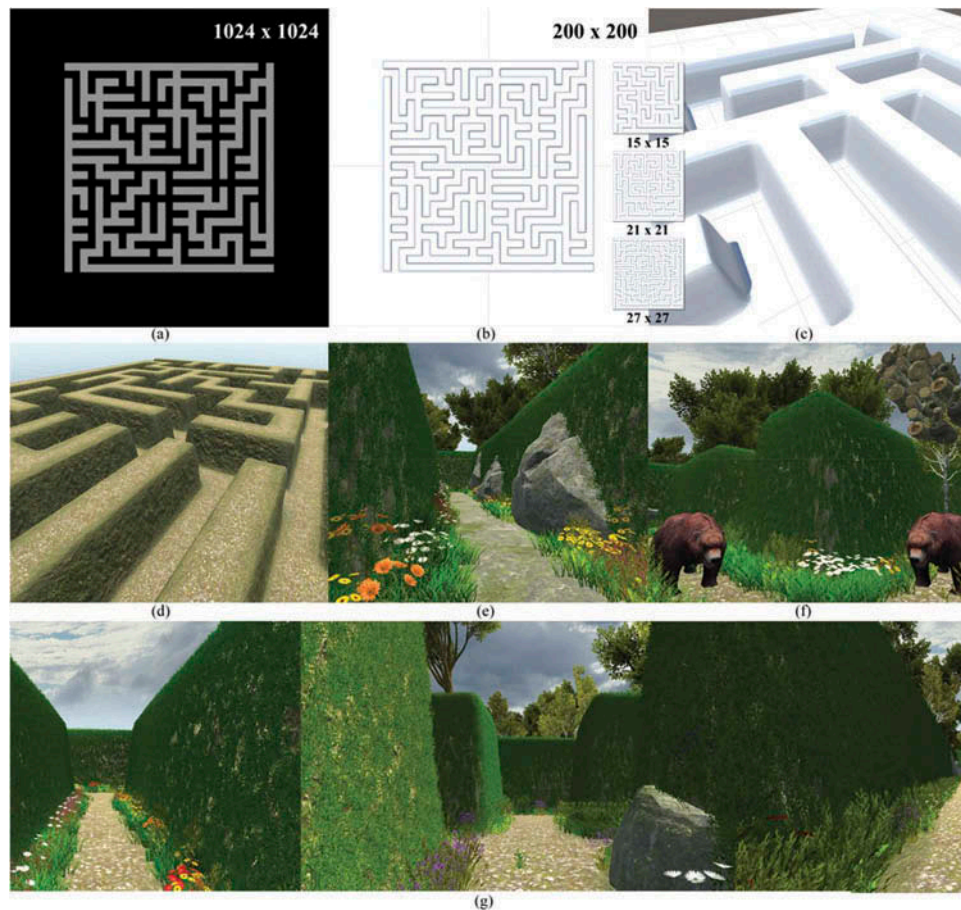


Figure 10. Mazemap-based MVR scene generation: (a) mazemap outcome, (b–c) mazemap-based generation of 3D terrain (top view, perspective view), (d) manual work on texture, (e) manual graphic work with grass, trees, and props, (f) arrangement of immersive events, (g) a generated MVR scene.

by users. Therefore, the height of a maze constructed from a noise map must be set in such a range that the surrounding paths and events are invisible. In a mazemap, the color 0

refers to the ground and 0.5 refers to a wall; the color of the noise map is multiplied to the maze wall to generate the combined results, as shown in Figure 11(b). Thus, the height

Table 1. The maze terrain setting of the MVR system.

Mazemap	Number of polygons	Number of events	Terrain production time	fps (frames per second)
15 × 15	1,517,124	9	6h	80.15
21 × 21	3,053,979	16	9h	76.75
27 × 27	6,021,282	38	15h	75.25

**Figure 11.** Result of a maze terrain with an irregular shape generated by combining the proposed mazemap with a noise map: (a) Gaussian noise map, (b) result of the combination of the noise map and the mazemap, (c) MVR scene with a natural look.

of the 3D-maze terrain should be set so that it is not less than the specified minimum range. The calculation process is as follows. If the maximum height of the maze wall is h , when the map color is 0.5, the height becomes h . In this study, the

minimum height of the maze (h_{min}) wall was set to half of the maximum height (h). Then, the mazemap color corresponding to the minimum height became 0.25 ($=0.5 \cdot h_{min}/h$). Therefore, if the minimum color of the noise map is c_{min} , the $c_{min}(= h_{min}/h)$ is $0.5 \cdot c_{min} = 0.25$, which should be calculated. Consequently, a noise map with a range of 0 ~ 1 is generated using Gaussian noise and the final noise map, as shown in Figure 11(a), is calculated by linearly interpolating the noise map in a range from c_{min} to 1.

The second experiment concerned the proposed immersive events, which were designed under two aspects: to enhance user immersion in a maze environment and to minimize VR sickness. We tested whether the intended effects of the designed events were actually experienced by the user in the MVR environment. As shown in Figure 12, each event ran optimally according to the given situation. It was important for the immersive events to run in an effective structure because both the situation and positioning were automatically generated based on maze pattern analysis.

The last experiment was conducted to check whether the VR environment of the proposed maze terrain delivered new experiences to users as intended by design. The experiment was divided into two parts. In the first part, a subjective survey on the psychological factors was carried out to analyze the adequacy and efficiency of the proposed maze scene and mazemap and the utilization of the proposed event. In the second part, a statistical survey was performed to analyze the presence, immersion of hand interaction, and VR sickness experienced by users in the maze space. The questionnaire was administered to 30 randomly selected participants between the ages of 21 and 35. They were exposed to the experimental setting, which was precisely configured according to the proposed maze terrain generation in a standing position using Oculus Rift DK2 HMD and Leap Motion (Figure 13). The experiment was conducted in a space of 1.5 m × 15 m, in which each participant could move through interactions using the hands, freely gazing around in a full 360° range.

5.1. Survey of the Psychological Factors of the Proposed Mazemap and Event

The first survey experiment was designed to analyze the psychological effects of the proposed mazemap and the virtual scenes and events created in this mazemap based on the user's experience. The study participants first engaged with the system to determine whether the maze environment adequately delivered new experiences. They then responded to four questionnaire items. The first questionnaire item concerned user satisfaction (the system would be deemed useless if users are not satisfied with the virtual maze terrain scenes). The satisfaction was measured at two levels: 1) satisfied or dissatisfied; 2) the degree of satisfaction on a five-point scale, with "3" being neutral. As shown in Figure 14, all participants answered that they were satisfied with the system and scored it with 4.25 points on average. Thus, it can be assumed that the proposed MVR system provided new experiences in a virtual environment.



Figure 12. Results of implementing the proposed immersive events: (a) process of immersion-enhancing events that were arranged automatically according to mazemap patterns, (b) sickness-reducing events that were operated according to the proposed conditions.



Figure 13. Experimental setting for experiencing the MVR system.

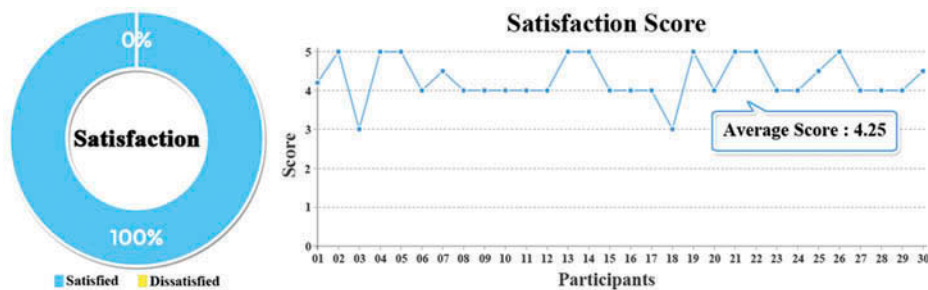


Figure 14. Experimental results concerning user satisfaction with the proposed MVR system.

The next experiment tested the continuity and efficiency of the mazemap, which automatically generates diverse maze patterns and offers users unlimited number of new maze environments, as earlier described. In contrast to a single-use terrain, this continuously renewed maze terrain is an important trait of the proposed MVR system. Therefore, we experimentally determined whether this continuity of novel experiences and diversity of maze patterns reduced boredom and enabled deeper immersion by continuously arousing the user's interest. The participants were asked whether they realized the maze pattern changes and to what extent (on a five-point scale) the various maze patterns influenced them in terms of arousing interest, enhancing immersion, and offering new experiences. As a result, it was determined that 83% of the

participants realized the maze pattern changes. The average scores for interest, immersion, and experience were 4.04, 4.11, and 3.90, respectively. In particular, mazemap obtained the highest score in immersion enhancement. This led us to assume that the maze-map-based maze terrain generation had positive effects on users by arousing interest and providing new experiences in a highly immersive environment (Figure 15).

We designed event elements tailored to a maze environment with the aim of enhancing user immersion, and we conducted experiments to test them. The effects of the proposed immersive events on the user were analyzed based on the participants' responses to three questionnaire items concerning new experiences in a VR environment: arousing

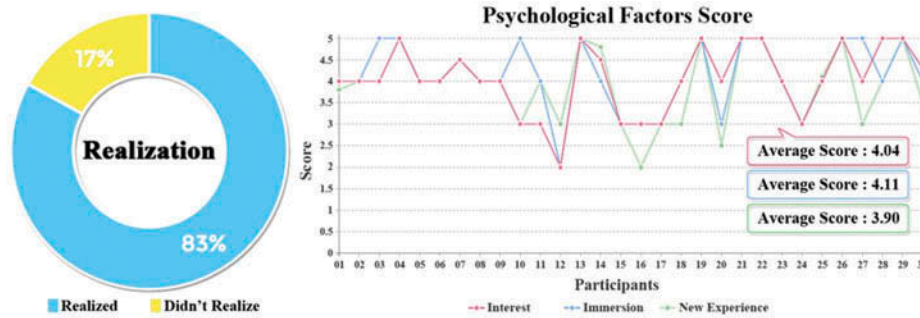


Figure 15. Analysis results of the experiment concerning the continuity and psychological factors of mazemap in relation to the diversity of its maze patterns.

interest, enhancing immersion, and providing new experiences. The participants were asked to rate each of them on a five-point scale, with higher scores given to more positive experiences. In the case of no event input, a neutral score of “3” was assigned. The experimental results revealed that immersion obtained the highest score (4.24 points), followed by interest (4.18 points), and new experience (3.93 points). It was thus confirmed that the proposed events fulfilled their intended functions by enhancing immersion and offering new experiences in VR settings for arousing user interests (Figure 16).

Sickness-reducing events were additionally analyzed. In a preliminary experiment, the minimum duration of the user remaining in a maze without feeling VR sickness was determined to be 90 seconds. For time-measuring purposes, we designed the events in three steps. First, we measured the time spent on escaping the maze depending on the size of the mazemap. We derived statistical data on the average time up to the point of incurring sickness in the maze environment. In the questionnaire, the participants were asked to respond in terms of whether the proposed events helped them reduce the escape time and to rate the satisfaction on a five-point scale. The experimental result revealed that the average times for escaping the mazemap maze patterns with sizes of 15×15 , 21×21 , and 27×27 were 78.9, 107.8, and 187.0 seconds, respectively. In the experiment of a maze without sickness-reducing events, 40% of the participants answered that they felt clearly sick while escaping the maze, and the analysis of the records revealed that the average time until the sickness

developed was approximately 120 seconds. In other words, the participants could rest 30 seconds longer in the maze without incurring sickness than the time derived in the preliminary experiment (90 seconds). In the experiment with sickness-reducing events, all of the participants answered that these events kept them from staying in the maze longer than optimal, and the average satisfaction score for the sickness-reducing effect of the proposed events was 4.15 (Figure 17). Thus, the proposed events were confirmed to reduce the negative factors of the maze environment.

5.2. Statistical Experiment on the Presence, Immersion of Hand Interaction, and VR Sickness

The second survey experiment was designed to systematically analyze whether the MVR provides a high presence to users and whether the proposed hand interaction causes VR sickness and provides higher immersion in comparison to conventional input devices, such as a keyboard. First, we conducted an experiment to determine whether the new VR composed of maze terrain provides a satisfactory presence for users. For this experiment, we used the presence questionnaire (Witmer, Jerome, & Singer, 2005) in the same environment as the one used for the experiment group in this study. This presence questionnaire consists of 24 items that can, generally, analyze the presence experienced by the user in a virtual space. Furthermore, to obtain a more detailed analysis of presence, the 24 items are classified as follows. Items 3, 4, 5, 6, 7, 10, and 13 can be used for a detailed analysis of realism;

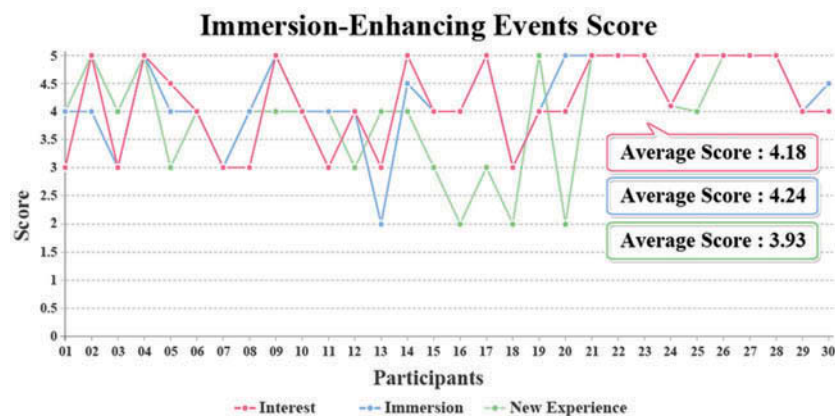


Figure 16. Experimental results concerning user reactions to the proposed immersion-enhancing events.

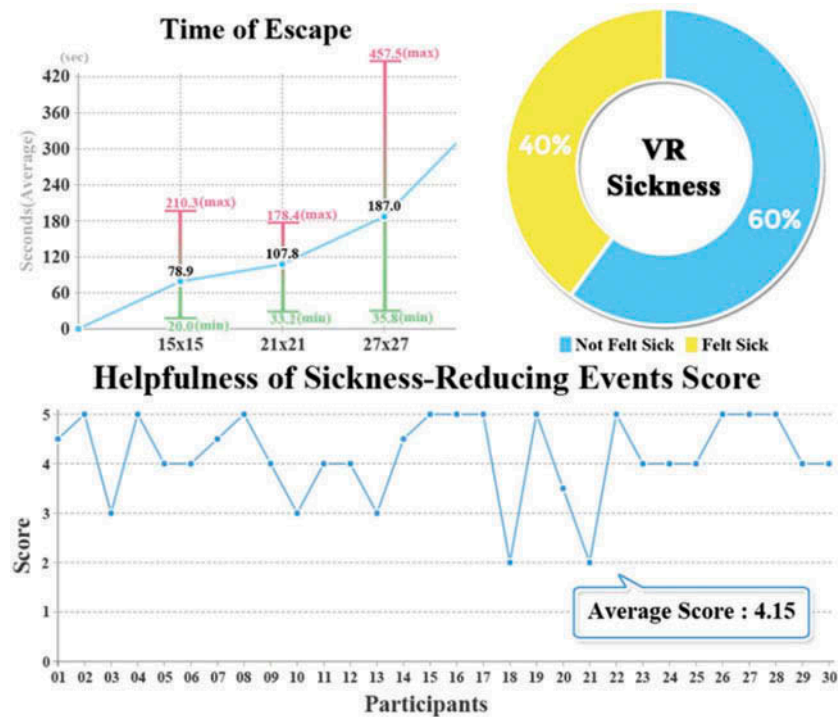


Figure 17. Analysis results of the experiment on VR sickness in a maze environment.

items 1, 2, 8, and 9 can be used to analyze the possibility to act; items 14, 17, and 18 can be used to analyze the quality of the interface; items 11, 12, and 19 can be used to analyze the possibility of examining the user's response; items 15 and 16 can be used for the user's self-evaluation of performance; items 20, 21, and 22 can be used to analyze the sounds; and items 23 and 24 can be used to analysis the haptic technology. Users were instructed to enter a score between 1 and 7 for each item. Only items 1–19 were used in this survey because the MVR of this study does not consider sounds and haptic technique. Table 2 outlines the results. The raw data obtained by dividing the total average score (118.7) by 19 items is 6.25, which is close to 7. This shows that the proposed MVR system offers a generally high presence. Furthermore, detailed items, such as realism and the possibility to act, also received high scores in general. In particular, the analysis of the quality of the interface showed that the proposed hand interaction helped improve the presence of the users.

Next, the same questionnaire was used to determine whether the proposed hand interaction provides a higher presence compared with conventional input devices, such as a keyboard. For this analysis, 10 items (2, 3, 5, 7, 8, 13, 14, 16, 18, and 19) were selected, which are appropriate for comparing interactions among the 19 items of the questionnaire. The

survey was conducted twice during the process of searching the same maze space: first using the keyboard and then using hand interaction. Based on the values entered by the participants, the Wilcoxon test was used to statistically analyze whether the proposed hand interaction method provides a better user experience than conventional input devices, such as a keyboard. The Wilcoxon test is a non-parametric statistical hypothesis test proposed by Wilcoxon (1945) to analyze the significance or insignificance of the difference between two related samples. After calculating the significance probability (p -value) using the Wilcoxon test, the significance probability of using a keyboard or hand interaction was 0.0045086984, which was smaller than the significance level (0.05). Thus, the null hypothesis was rejected. In other words, this result proved that motion control using the proposed hand interaction provides a better presence than a conventional input device.

The last step of the second statistical experiment is an analysis of VR sickness. The results of the preceding experiment, shown in Figure 16, confirm that the proposed event is helpful for preventing VR sickness. However, if the MVR system users basically feel VR sickness, they may experience problems when trying to apply the system to VR. Therefore, a survey was conducted to statistically analyze the VR sickness of the MVR system. Reliable results were obtained using the simulator sickness questionnaire (SSQ) (Kennedy, Lane, Berbaum, & Lilienthal, 1993). Similar to the presence questionnaire, the SSQ is used to systematically analyze VR sickness through a 16-item questionnaire. In addition, the items are classified to enable analysis of sickness separately for nausea, oculomotor, and disorientation. For the objective analysis of symptoms, raw data, excluding weight, is sometimes used (Bouchard, St-Jacques, Renaud, &

Table 2. Analysis results of the presence questionnaire for the MVR system.

	Mean	Standard deviation (SD)
Total	118.4	8.35
< realism >	43.4	3.15
< possibility of act >	25.4	1.85
< quality of interface >	17.9	1.81
< possibility to examine >	19.3	1.68
< self-evaluation of performance >	12.4	1.36

Table 3. Analysis of the simulator sickness questionnaire (SSQ) results for the proposed MVR system.

	Mean	SD	Minimum	Maximum
Original SSQ(weighted)				
Total	31.538	24.471	0	82.28
Nausea	14.31	13.988	0	47.7
Oculomotor	25.772	19.532	0	60.64
Disorientation	52.896	46.71	0	167.04
Alternative SSQ(raw data)				
Total	6.50	5.21	0	19
Nausea	1.50	0.09	0	5
Oculomotor	3.4	2.58	0	8
Disorientation	3.8	3.36	0	12
After 120 sec	7.10	5.85	0	24

Wiederhold 2009). In this study, we obtained and analyzed the results from both methods. Table 3 shows the SSQ results of the MVR system when hand interaction was used. The results of the survey showed that the VR sickness of the MVR system was slight or less. Based on the fact that that the motion sickness time in the maze was measured at about 120 seconds in the preceding experiment, the VR sickness values before and after 120 seconds were compared and analyzed. Thus, the VR sickness was found to increase, although the difference was not large. This difference could be reduced by reducing the maze escape time through the proposed sickness-reducing events.

6. Limitations and Discussion

In this study, the maze provided by the MVR system has a limitation in that it can only generate square-shaped patterns. We did not consider the diversity of maze patterns because we wanted to focus on whether the VR composed of maze terrains can provide users with a new experience and presence. Nevertheless, some studies have been conducted to produce art images with maze patterns, including circular patterns. In particular, Xu and Kaplan (Xu & Kaplan, 2007) proposed an algorithm to generate diverse unique maze patterns based on images, although they were not 3D maze spaces. Their algorithm generates a unique mazemap based on the contours of input images for a maze with clear start and end points. Therefore, we expect that the MVR system can be expanded to provide users with a new experience by combining the artistic maze generation algorithm with the mazemap used in this present study. New experiences with the maze can be provided to users by applying reconfigurable mazes to the MVR system using the impossible figure proposed by Lai et al. (Lai et al., 2016). The maze generated from the impossible figure can provide various path-finding experiences, even if the maze terrain is small. If the immersion-enhancing events proposed in this study are applied, they could be used as a new MVR to enhance the immersion experience of users. Clearly presenting the start and end with a guiding principle could present an alternative solution to reducing VR sickness. However, because the user is watching the same terrain repeatedly, a new background should be created with a seamless maze so that the user feels less trapped in a small VR space, which is a major cause of VR sickness. Finally, further research could be conducted on the maze generation interface based on sketches that have not been studied previously.

7. Conclusion

In this article, we proposed the MVR system, which generates maze-based virtual environments consisting of diverse events and interactions to provide the user with new experiences in an immersive VR setting. The MVR system was designed with two key components: mazemap, which automatically generates 3D maze terrains with diverse maze patterns, and manual processing of immersive events that are tailored to maze settings. Accordingly, immersion is enhanced and sickness is minimized with increased interaction. We first presented the user-perspective finite maze generation algorithm based on Prim's algorithm for computing multi-patterned mazes with finite paths. Then, we presented our designed mazemap using a heightmap to implement realistic expressions of maze patterns in a 3D VR space. Mazemap enables the automated generation of textures to reflect any given maze pattern and height information. It provides the user with new maze environment experiences from start to end points in a finite maze computed via a five-step procedure. The mazemap-generated 3D maze terrain shape then undergoes graphical post-processing for increased realism. To enhance the immersive experience in the MVR system, we additionally presented immersive events and interactions. We designed them to be automatically generated in a manner that enhances user immersion and minimizes sickness while the user experiences the maze. Immersion-enhancing events were designed to induce the user's tension and concentration to align with each maze-map-generated maze pattern. Sickness-reducing events were designed to suit the purpose of preparing the user for situations that can occur in case of prolonged confinement in the maze. These events were manually placed to ensure accuracy because of their direct relation to sickness effects. Finally, we designed Leap Motion-based interactions to facilitate relatively free movements and motions for object control by reducing spatial constraints in the VR environment. The interaction patterns were designed to follow four different hand motions to enable the user to freely move and control objects in the maze. To test whether the MVR system provides new experiences for actual users, we then conducted various questionnaire-based experiments. Most of the participants answered that the MVR system delivered new experiences, and the continuously renewed mazemap-generated maze patterns ensured the continuity of the maze environment. The proposed immersive events were found to enhance user satisfaction in terms of inducing interest, maintaining immersion, and providing new VR experiences. In the analysis of sickness that is likely to occur in a complex maze space, the average time before incurring sickness due to fatigue and dizziness by remaining too long in a maze was found to be 120 seconds. The events designed in the MVR system to prevent sickness were determined to be helpful in mitigating sickness.

The MVR system thus enables the fully automated generation of various virtual maze scenes to provide users with new experiences. However, a certain portion of the system requires manual processing on account of the nature of the system. Furthermore, the number of immersive events is limited because we focused on the automatic arrangement of events to match the maze pattern rather than on varying their types.

In further research, these issues will be considered and efforts will be made to optimize the MVR system by minimizing manual processing and designing various events. The limitation of the square-shaped patterns of the mazemap-generated maze will also be addressed and the generation of various shapes will be made possible. Future research will additionally involve interactive maze escape experiences in a high-immersion environment by using a treadmill-type motion device, such as Virtuix Omni.

Funding

This research was supported by Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education (No. NRF-2014R1A1A2055834).

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